Day - 1

07 April 2025 09:35

1. Splunk:-
- Bucketing.

Inverted Monitoring.
- ELK
- Nagios.
- Datadog
- Dynatrace

→ Alert, Report, Dashboard, Graph.

Late 2000s → Data Storage → Data Warehouse / Data lake → Analysis

Splunk → Log Monitoring Tool

Source (App) → Splunk/ELK/Nagios → GUI, S1

Component of splunk:-

1. Forwarder:-

App logs, Syslogs, various security logs. → Forwarder (Uni) → Splunk.

PaloAlto, Fireeye

Forwarding Agent: create pipeline for data transfer b/w source & splunk end

2. Indexer:-

Source (Forwarder) → Splunk (Indexer) → Database (where we store the data in splunk).

3. Search Head:-

Source (Forwarder) → Indexer ⇄ Search Head (GUI) ⇄ User

SPL (Search Processing language)

Dashboard, Alert, Report, KO

4. Forwarder
- → Universal Forwarder (UF)
- → Heavy forwarder (HF)

Source (Forwarder) → Indexer

Parsing Phase.

1. Parsing on Indexer end:-

S1, S2, S3, ..., S1000 → Indexer → Indexing Phase.

12:45 PM

Parsing 45 min. Indexing Delay

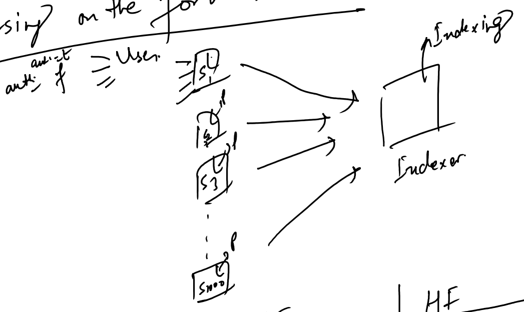
12 PM

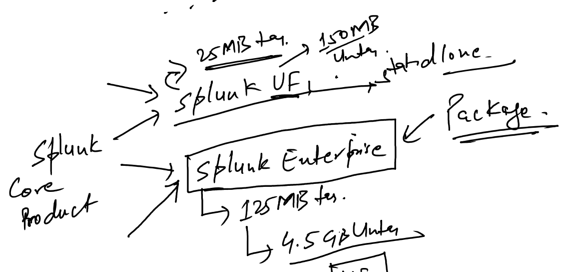
Parsing
- → In-built Parsing.
- → Custom Parsing.

12PM ← Indexing

Parsing → Custom Parsing

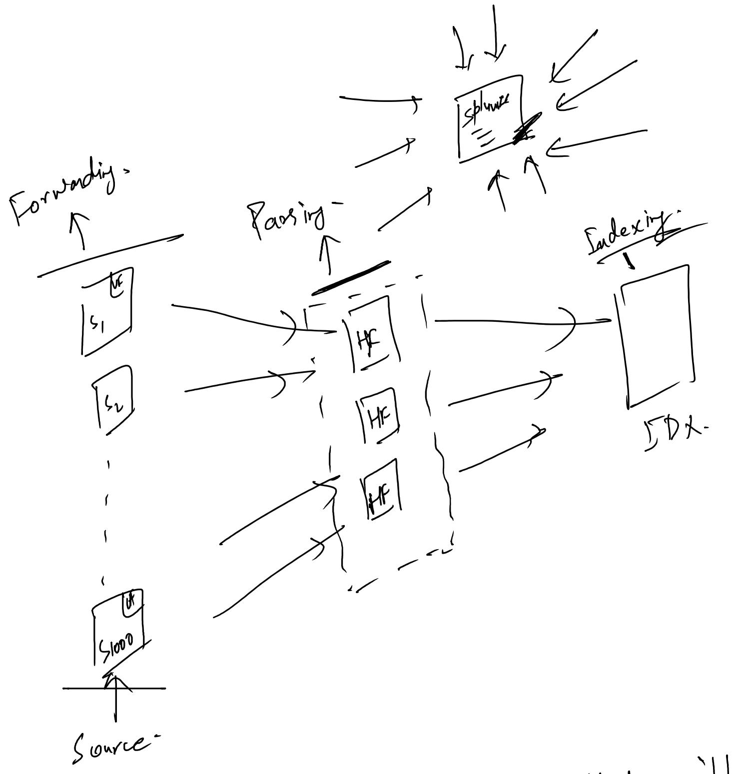
2. Parsing on the forwarder end:





UF	HF
1. Splunk UF	1. Splunk Enterprise
2. Parsing ↳ Indexer	2. HF ↳ Forwarder ↳ Indexer

(SH, Indexer, HF, LM)



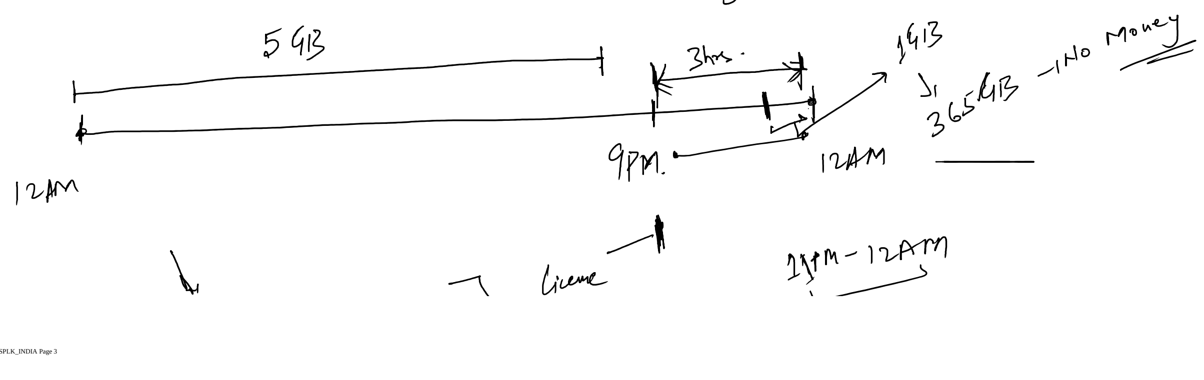
4. License Master Policing Agent that will verify whether the Splunk in your license Agreement.

↳ Amount of Data, you ingest on the daily basis.

5GB/d → 1year.

↓

24hrs Cycle.



Indexing will continue
Searching will be stopped

License Breach

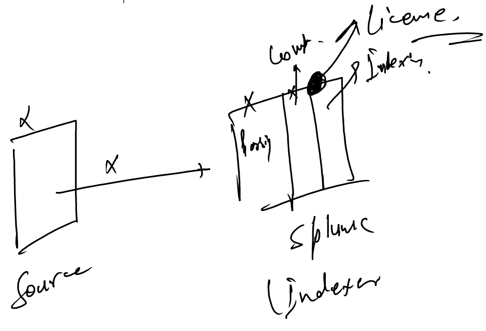
11PM - 12AM

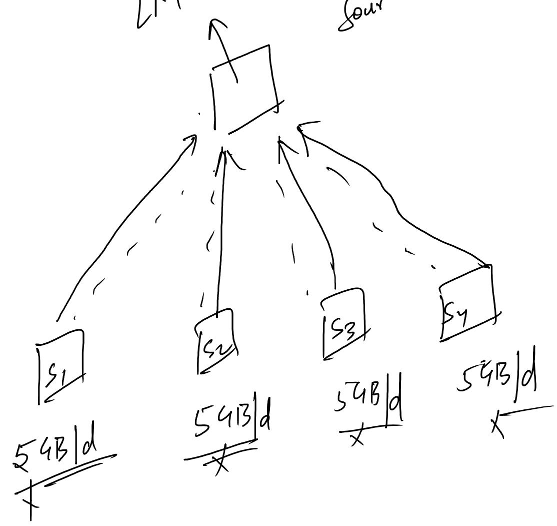
1GB

5 times - 30 day window.

Splunk blacklist.

* License Pooling:-



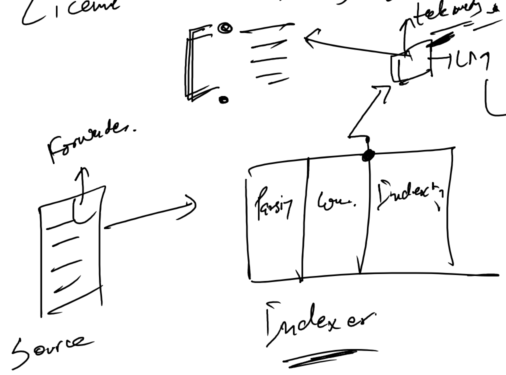


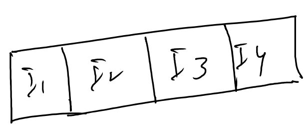
(1) upper limit → 5GB/d

$S_1 + S_2 + S_3 + S_4 \not> 20GB$

License Master is down ? → Timer 72 hours.

Serv connect with LM



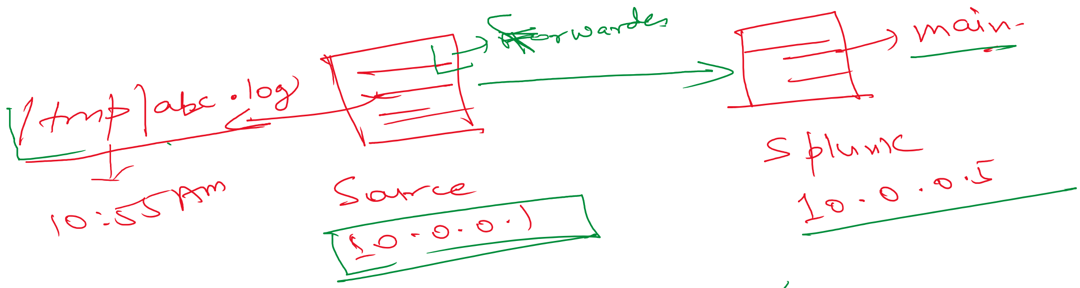


Indexer

Predefined. Indexes
→ Reserved for the internal logs of the splunk Application.
→ Ex → _* → _internal, _audit, _introspection etc.
→ No license Consumed.
→ Can't Ingest custom data.
... name main.
data.

Index
- → Predefined. Indexes
  - → ex, _
  - → No License Con..
  - → Can't Ingest custom. main.
- → Default Index
  - → Index with name
  - → License Consumed.
  - → Can Ingest Custom data.
- → Custom Index
  - → Create index with custom name.
    ex: VK-idx.
  - → License Consumed.
  - → Can Ingest the data.

Forwarder. Index



Host → 10.0.0.1

Source → /tmp/abc.log

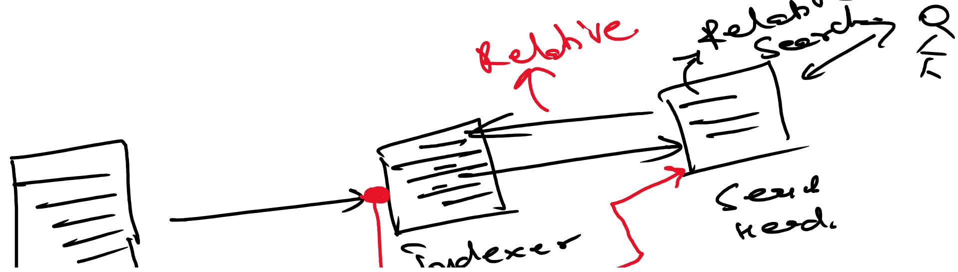
datatype ← Sourcetype → log

index → main

_time → 10:55 AM

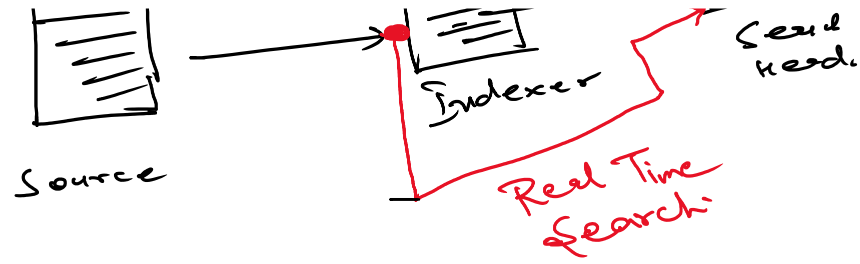
Search Mode
- → Fast Mode (Min). – events, No field extra
- → Smart mode – events + fields
- → Verbose mode (Max) – event + fields +

index=_internal
- → (1) event is pulled.
- → (2) field is extracted.



ction.

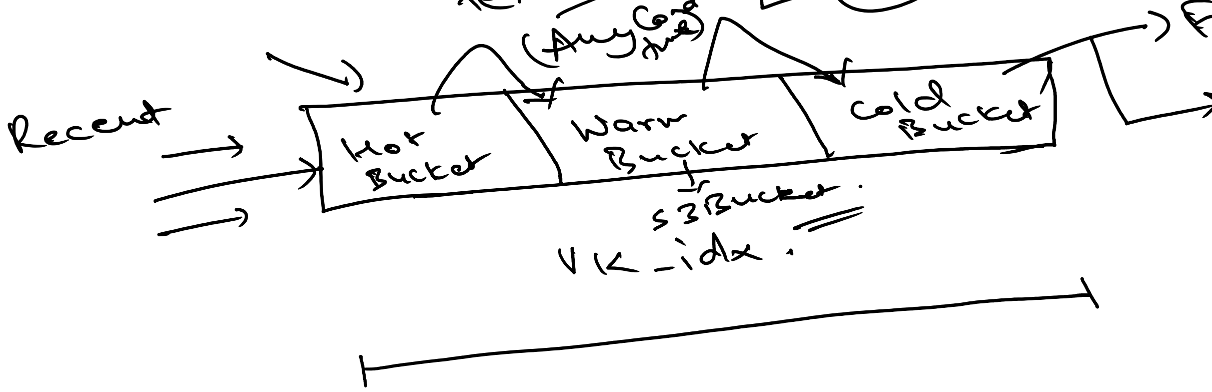
Navigate b/w
tabs



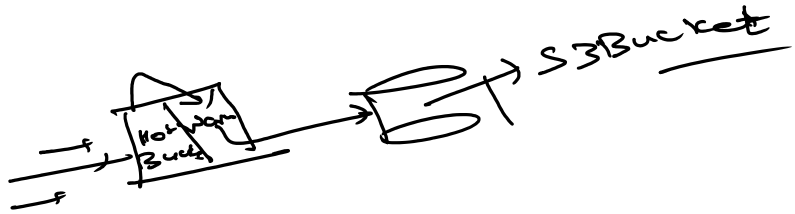
Buckets:-

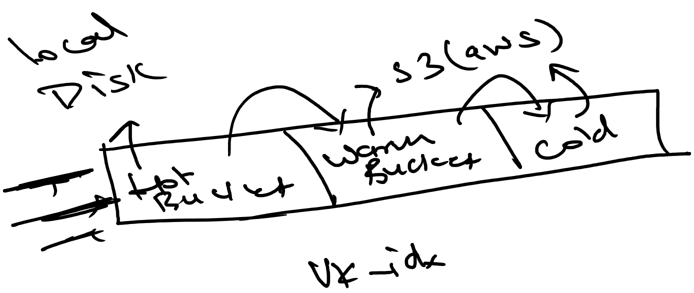
1. Hot Bucket.
2. Warm Bucket.
3. Cold Bucket.
4. Frozen Bucket
5. Thawed Bucket

Retention Period. ←
1. Age of Data ≥ 7d
2. Size of B



Smart Store:-





SPL Commands:-

1. table → Tabular output. Syn:- |table f1, f2, f3
2. rename. → search time change in the name of
3. stats →
4. eval.

t

levent 2000 45

ucket

rozen
Bucket
Thrawed
Bucket

fieldName

3. stats
4. eval.
5. Top
6. rare.
7. sort
8. dedup

Field Name is Case sensitive & Field Value

Case insensitive

function

Stats → 1. count —
2. sum — Summation of the numeric value
3. Avg.
4. list — grouping purpose
5. Values —

Statistical data

A	B	C	D
	—		—
—	—	NA	
—	—		—

fillnull value="NA"

Eval command :—

Evaluation Command.

a = b + c

eval - initialize the Varia

variable

(int, str, var)

eval a = b + c

dis
ble .

if-else statement :-

True

if (a>b, a, b)

Condition. False

```
if (a>b)
  {
    print(a);
  }

else {
    print(b);
  }
```

Sort → Sort (f) → Ascending

Default. Sort → Descending.

Case statement :-

```
switch
  case (a): ____
  case (b): ____
  case (c): ____
  case (d): ____
  default (1): ____
```

Case (cond1, "___", cond2

1=1, "___")

Case statement

, "——", cond, "——",

works.